Metadata	Nilai
Dokumen	Complete API Contract Collection & Response Schemas
Proyek	Real-Time Auction & Bidding Engine (Fintech / E-Commerce)
Versi	v2.0.0 (Corporate Baseline)
Status	`Approved Baseline`
Tanggal	2026-10-24
Base URL	`http://localhost:8080/api/v1`

1. Standar Format Respon Global

Respon Sukses (HTTP 200 / 201)

```
{
  "status": "success",
  "data": {}
}
```

Respon Gagal (HTTP 400 / 401 / 403 / 404 / 409 / 500)

```
{
  "status": "error",
  "error": {
    "code": "ERROR_CODE_STRING",
    "message": "Pesan deskripsi kesalahan dalam bahasa Indonesia."
  }
}
```

2. Koleksi Endpoint Lengkap Beserta Seluruh Respon

MODUL 1: AUTENTIKASI DUAL-LOGIN & RBAC

1. `GET /auth/niro/authorize-url`

Frontend (Next.js) memanggil endpoint ini ketika pengguna mengklik tombol *"Login with NIRO"*. Endpoint ini mengembalikan URL redirect otorisasi SSO NIRO beserta state parameter untuk keamanan CSRF.

- **Headers:** *None*
- **Response 200 OK:**

```json

{

"status": "success",

"data": {

"redirect_url": "https://auth.niro.internal/oauth2/authorize?
client_id=salt_auction_web&response_type=code&redirect_uri=http%3A%2F%2Flocalhost%3A3000%2Fauth%2Fcallback&state=sec_state_981247",

"state": "sec_state_981247"

}

}

```

2. `POST /auth/login/niro`

Setelah Frontend membuka URL redirect NIRO, pengguna berhasil login di sistem NIRO dan FE mendapatkan token/kode NIRO. Frontend kemudian mengirimkan token NIRO tersebut ke Backend melalui endpoint ini.

Logika Pemeriksaan Akun di Backend:

- Backend memvalidasi token ke server NIRO dan mengekstrak identitas (`niro_user_id`, `email`, `username`).
- Backend memeriksa tabel `users` berdasarkan `niro_user_id`:

- **Jika Belum Terdaftar:** Sistem secara otomatis **mendaftarkan akun baru** (`auth_provider = 'NIRO'`, role default `BIDDER`), membuatkan akun dompet (`available_balance = 0`, `locked_balance = 0`), dan menerbitkan token JWT sesi. Respon mengembalikan status `is_new_user: true` (HTTP 201).

- **Jika Sudah Terdaftar:** Sistem langsung **meloginkan pengguna**, memuat role terkini (`BIDDER` atau `ADMIN` jika sudah dipromosikan), dan menerbitkan token JWT sesi. Respon mengembalikan status `is_new_user: false` (HTTP 200).

- **Headers:** `Content-Type: application/json`
- **Request Body:**

```json
{
"niro_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9.niro_sample_token..."
}
```

- **Response 200 OK (Pengguna Sudah Terdaftar — Berhasil Loginkan):**

```json
{
"status": "success",
"data": {
"is_new_user": false,
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.auction_jwt_token...",
"user": {
"id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
"niro_user_id": "NIRO-USR-99812",
"username": "budi_kolektor",
"email": "budi@collector.id",
"auth_provider": "NIRO",
"role": "BIDDER"
},
"wallet": {
"available_balance": 150000000.00,
"locked_balance": 0.00,
"total_balance": 150000000.00
}
}
}
```

- **Response 201 Created (Pengguna Belum Terdaftar — Berhasil Didaftarkan Otomatis):**

```json
{
"status": "success",
"data": {
```

"is_new_user": true,
"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.auction_jwt_token...",
"user": {
"id": "7ca12345-5717-4562-b3fc-2c963f66bbbb",
"niro_user_id": "NIRO-USR-10293",
"username": "siti_investor",
"email": "siti@luxury.com",
"auth_provider": "NIRO",
"role": "BIDDER"
},
"wallet": {
"available_balance": 0.00,
"locked_balance": 0.00,
"total_balance": 0.00
}
}
}
```

- **Response 401 Unauthorized (Token NIRO Invalid/Expired):**

```json
{
"status": "error",
"error": {
"code": "ERR_INVALID_NIRO_TOKEN",
"message": "Token otentikasi NIRO tidak valid atau telah kedaluwarsa."
}
}
```

---

3. `POST /auth/login/email`

Login terisolasi khusus Superadmin platform lelang menggunakan Email dan Password internal (minimal 12 karakter), dilengkapi proteksi penguncian bertingkat (*Progressive Lockout*).

Aturan Kebijakan Gagal Login (Lockout):

- 3x salah pertama: Terkunci **1 menit**.
- 3x salah kedua (total 6x): Terkunci **2 menit** (dobel waktu).
- 3x salah ketiga (total 9x): Terkunci **1 jam (60 menit)**.
- Catatan: Fitur *Forgot Password* dan *Kickout* ditiadakan.
- **Headers:** `Content-Type: application/json`
- **Request Body:**

```json
{
"email": "superadmin@salt-auction.internal",
"password": "MasterAdminSalt2026!"

}

```

- **Response 200 OK (Login Berhasil):**

```json

{

"status": "success",

"data": {

"token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.superadmin_jwt...",

"user": {

"id": "00000000-0000-0000-0000-000000000001",

"username": "superadmin",

"email": "superadmin@salt-auction.internal",

"auth_provider": "INTERNAL",

"role": "SUPERADMIN"

}

}

}

```

- **Response 401 Unauthorized (Kredensial Salah):**

```json

{

"status": "error",

"error": {

"code": "ERR_INVALID_CREDENTIALS",

"message": "Email atau kata sandi salah.",

"details": {

"remaining_attempts_before_lock": 2

}

}

}

```

- **Response 429 Too Many Requests (Akun Terkunci Sementara - Lockout):**

```json

{

"status": "error",

"error": {

"code": "ERR_ACCOUNT_TEMPORARILY_LOCKED",

"message": "Terlalu banyak percobaan login gagal berturut-turut. Akun Anda dikunci sementara selama 60 detik. Silakan coba lagi nanti.",

"details": {

"retry_after_seconds": 60,

"lockout_level": 1

}

}

}

```

---

3. `GET /admin/users`

Mengambil daftar pengguna login NIRO untuk dikelola oleh Superadmin.

- **Headers:** `Authorization: Bearer `
- **Query Params:** `?page=1&limit=10&role=BIDDER`
- **Response 200 OK:**

```json

{

"status": "success",

"data": {

"users": [

{

"id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",

"niro_user_id": "NIRO-USR-99812",

"username": "budi_kolektor",

"email": "budi@collector.id",

"auth_provider": "NIRO",

"role": "BIDDER",

"created_at": "2026-10-24T10:00:00.000Z"

}

],

"pagination": {

"page": 1,

"limit": 10,

"total_items": 1,

"total_pages": 1

}

}

}

```

- **Response 403 Forbidden:**

```json

{

"status": "error",

"error": {

"code": "ERR_FORBIDDEN_ACCESS",

"message": "Akses ditolak. Endpoint ini hanya dapat diakses oleh Superadmin."

}

}

```

```

4. `PATCH /admin/users/{id}/role`

Superadmin mempromosikan pengguna login NIRO menjadi **ADMIN** (Auction Manager).

- **Headers:** `Authorization: Bearer `, `Content-Type: application/json`
- **Request Body:**

```json
{
"role": "ADMIN"
}
```

- **Response 200 OK:**

```json
{
"status": "success",
"data": {
"id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
"username": "budi_kolektor",
"previous_role": "BIDDER",
"new_role": "ADMIN",
"updated_at": "2026-10-24T12:05:00.000Z"
}
}
```

- **Response 404 Not Found:**

```json
{
"status": "error",
"error": {
"code": "ERR_USER_NOT_FOUND",
"message": "Pengguna dengan ID tersebut tidak ditemukan."
}
}
```

MODUL 2: DOMPET LIKUIDITAS & TOP-UP SALDO

5. `GET /wallet/me`

Mengambil informasi saldo pengguna saat ini.

- **Headers:** `Authorization: Bearer `
- **Response 200 OK:**

```json
{
"status": "success",
"data": {
```

"user_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
"available_balance": 25000000.00,
"locked_balance": 75000000.00,
"total_balance": 100000000.00
}
}
```

---

**6. `POST /wallet/topup`**

Menambahkan saldo bebas (*Available Balance*) untuk modal bidding (*Role BIDDER Required*).

Sesuai aturan bisnis **US-001**, **hanya Bidder yang dapat melakukan top-up**, dan nominal minimal pengisian adalah **Rp 100.000**.

- **Headers:** `Authorization: Bearer `, `Content-Type: application/json`
- **Request Body:**

```json
{
"amount": 50000000.00
}
```

- **Response 200 OK:**

```json
{
"status": "success",
"data": {
"topup_amount": 50000000.00,
"previous_available_balance": 25000000.00,
"new_available_balance": 75000000.00,
"locked_balance": 75000000.00,
"total_balance": 150000000.00
}
}
```

- **Response 400 Bad Request (Nominal di Bawah Minimal Rp 100.000):**

```json
{
"status": "error",
"error": {
"code": "ERR_MINIMUM_TOPUP_AMOUNT",
"message": "Minimum top up is IDR 100,000.",
"details": {
"minimum_required": 100000.00
}
}
}

```
```

- **Response 403 Forbidden (Bukan Bidder):**

```json
{
"status": "error",
"error": {
"code": "ERR_FORBIDDEN_ROLE",
"message": "Akses ditolak. Fitur dompet dan top-up hanya dapat diakses oleh peran Bidder."
}
}
```

"locked_balance": 75000000.00,
"total_balance": 150000000.00
}
}
```

- **Response 400 Bad Request:**

```json
{
"status": "error",
"error": {
"code": "ERR_INVALID_TOPUP_AMOUNT",
"message": "Nominal top-up harus berupa angka positif lebih dari nol."
}
}
```

MODUL 3: KATALOG LELANG BARANG MEWAH

7. `GET /auctions/active`

Mengambil daftar snapshot seluruh lot barang bernilai tinggi yang sedang aktif di pasar, mendukung pagination.

Gambar aset di-return dalam format **Base64 Compressed String** (hasil decode dari binary BLOB di DB).

- **Query Params:** `?page=1&limit=10&category=LUXURY_WATCHES`
- **Response 200 OK:**

```json
{
"status": "success",
"data": {
"auctions": [
{
"id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
"title": "Cosmograph Chronograph Steel 2023",
"category": "LUXURY_WATCHES",
"condition_grade": "MINT",

"authenticity_guarantee": "Full Set 2023, Box & Papers",
"image_base64_compressed": "...",
"model_type": "CAPPED",
"status": "ACTIVE",
"starting_price": 350000000.00,
"current_price": 380000000.00,
"min_increment": 5000000.00,
"allowed_increment_steps": [5000000.00, 10000000.00, 25000000.00],
"valuation_ceiling": 500000000.00,
"highest_bidder_username": "andi_collector",
"end_time": "2026-10-25T12:00:00.000Z"
}
],
"pagination": {
"page": 1,
"limit": 10,
"total_items": 1,
"total_pages": 1
}
}
}
```

---

8. `GET /auctions/{id}`

Mengambil detail spesifikasi satu lot barang bernilai tinggi beserta riwayat penawaran terakhir.

- **Response 200 OK:**

```json
{
"status": "success",
"data": {
"lot": {
"id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
"title": "Cosmograph Chronograph Steel 2023",
"category": "LUXURY_WATCHES",
"condition_grade": "MINT",
"authenticity_guarantee": "Full Set 2023, Box & Papers",
"description": "White dial, pristine condition, unworn.",
"image_base64_compressed": "...",
"model_type": "CAPPED",
"status": "ACTIVE",
"starting_price": 350000000.00,
"current_price": 380000000.00,
"allowed_increment_steps": [5000000.00, 10000000.00, 25000000.00],

"valuation_ceiling": 500000000.00,

"highest_bidder_id": "8bb30e92-3a52-47d1-ba76-218f2d5f22e1",

"highest_bidder_username": "andi_collector",

"start_time": "2026-10-24T12:00:00.000Z",

"end_time": "2026-10-25T12:00:00.000Z"

},

"recent_bids": [

{

"bid_id": "7fa12345-872b-42e7-91a6-f7615cb38421",

"bidder_username": "andi_collector",

"amount": 380000000.00,

"created_at": "2026-10-24T12:15:30.000Z"

}

]

}

}

```

- **Response 404 Not Found:**

```json

{

"status": "error",

"error": {

"code": "ERR_AUCTION_NOT_FOUND",

"message": "Lot lelang dengan ID tersebut tidak ditemukan."

}

}

```

MODUL 4: PENAWARAN INKREMENTAL & SALDO TERTAHAN (HOT-PATH)

9. `POST /auctions/{id}/bids`

Mengajukan tawaran lelang dengan memilih tombol inkremental resmi (atau menawar langsung nilai ceiling).

Sistem secara otomatis mengevaluasi:

- Jika `current_price + increment_amount >= valuation_ceiling`, lelang **LANGSUNG LAKU (*Instant Buyout*)** seketika tanpa perlu flag manual.
- **Headers:** `Authorization: Bearer `, `Content-Type: application/json`
- **Request Body:**

```json

{

"increment_amount": 10000000.00

}

```

- **Response 200 OK (Tawaran Sukses & Memimpin):**

```json

{

"status": "success",
"data": {
"auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
"new_current_price": 390000000.00,
"bid_status": "WINNING",
"wallet_state": {
"locked_collateral_for_this_bid": 390000000.00,
"remaining_available_balance": 10000000.00,
"total_balance": 400000000.00
},
"time_extended": false,
"new_end_time": "2026-10-25T12:00:00.000Z",
"buyout_triggered": false
}
}

```

- **Response 200 OK (Mencapai/Melebihi Ceiling — Instant Buyout Terpicu Otomatis):**

```json
{
"status": "success",
"data": {
"auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
"new_current_price": 500000000.00,
"bid_status": "WINNING",
"wallet_state": {
"locked_collateral_for_this_bid": 500000000.00,
"remaining_available_balance": 50000000.00,
"total_balance": 550000000.00
},
"time_extended": false,
"buyout_triggered": true,
"auction_status": "CLOSED_AWARDED"
}
}
```

- **Response 400 Bad Request (Saldo Kurang):**

```json
{
"status": "error",
"error": {
"code": "ERR_INSUFFICIENT_AVAILABLE_BALANCE",
"message": "Saldo bebas Anda (Rp 25.000.000) tidak mencukupi untuk tawaran sebesar Rp 390.000.000. Silakan lakukan Top-Up."
}

}

```

- **Response 400 Bad Request (Lelang Sudah Ditutup):**

```json

{

"status": "error",

"error": {

"code": "ERR_AUCTION_NOT_ACTIVE",

"message": "Lelang ini telah berakhir atau telah dimenangkan via Buyout."

}

}

```

MODUL 5: BOT PENAWARAN OTOMATIS (AUTOMATED PROXY BIDDING)

10. `POST /auctions/{id}/proxy-bid`

Mendaftarkan atau memperbarui batas maksimal bot proxy bidding.

- **Headers:** `Authorization: Bearer `, `Content-Type: application/json`
- **Request Body:**

```json

{

"max_limit": 450000000.00

}

```

- **Response 200 OK:**

```json

{

"status": "success",

"data": {

"auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",

"proxy_id": "b1827461-872b-42e7-91a6-f7615cb38421",

"max_limit": 450000000.00,

"is_active": true,

"current_auction_price": 395000000.00,

"your_current_bid": 395000000.00,

"status": "PROXY_ACTIVE_AND_LEADING"

}

}

```

- **Response 400 Bad Request (Limit Lebih Kecil dari Harga Saat Ini):**

```json

{

"status": "error",

"error": {

"code": "ERR_PROXY_LIMIT_TOO_LOW",

"message": "Batas limit proxy harus lebih besar dari penawaran saat ini (Rp 390.000.000)."

}

}

```

---

11. `DELETE /auctions/{id}/proxy-bid`

Membatalkan / menonaktifkan aturan bot proxy bidding aktif.

- **Headers:** `Authorization: Bearer `
- **Response 200 OK:**

```json

{

"status": "success",

"data": {

"auction_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",

"is_active": false

},

"message": "Aturan proxy bidding berhasil dinonaktifkan."

}

```

MODUL 6: MANAJEMEN LOT & KURASI ADMIN

12. `POST /admin/auctions`

Admin / Auction Manager membuat listing lelang barang bernilai tinggi baru.

- Endpoint ini menerima payload dalam format **`multipart/form-data`** karena menyertakan unggahan binary berkas gambar aset (`image`).
- Gambar binary yang diunggah langsung disimpan di PostgreSQL sebagai **BLOB (`BYTEA`)** dan dikompresi menjadi format Base64 saat dibaca kembali.
- Pengaturan perpanjangan waktu anti-sniping (`anti_sniping_extension_minutes`) dikonfigurasi langsung pada endpoint ini.
- **Headers:** `Authorization: Bearer `, `Content-Type: multipart/form-data`
- **Form Data Fields:**

- `title` (text): `"Togo Leather Handbag 25 Gold Hardware"`

- `category` (text): `"DESIGNER_BAGS"`

- `condition_grade` (text): `"MINT"`

- `authenticity_guarantee` (text): `"Store Receipt 2024, Original Orange Box, Dustbag, CITES cert"`

- `description` (text): `"Store fresh condition, highly sought-after collector investment piece."`

- `image` (binary file): `[File binary gambar JPEG/PNG/WEBP]`

- `min_price` (number): `250000000.00`

- `allowed_increment_steps` (json string): `"[5000000, 10000000, 20000000]"`

- `max_price` (number, opsional): `400000000.00`

- `window_duration_seconds` (number): `86400`

- `anti_sniping_extension_minutes` (number, opsional): `2`

- **Response 201 Created:**

```json

{
"status": "success",
"data": {
"id": "c1f76d22-192a-4c28-8742-1928374aef12",
"title": "Togo Leather Handbag 25 Gold Hardware",
"category": "DESIGNER_BAGS",
"condition_grade": "MINT",
"authenticity_guarantee": "Store Receipt 2024, Original Orange Box, Dustbag, CITES cert",
"image_base64_compressed": "...",
"model_type": "CAPPED",
"status": "ACTIVE",
"current_price": 250000000.00,
"valuation_ceiling": 400000000.00,
"anti_sniping_extension_minutes": 2,
"start_time": "2026-10-24T12:30:00.000Z",
"end_time": "2026-10-25T12:30:00.000Z"
}
}
```

- **Response 400 Bad Request:**

```json
{
"status": "error",
"error": {
"code": "ERR_INVALID_AUCTION_PARAMS",
"message": "Harga maksimal harus bernilai lebih besar daripada harga minimal, dan file gambar wajib dilampirkan."
}
}
```

---

3. Spesifikasi Protokol WebSocket Lengkap

- **URL Koneksi:** `ws://localhost:8080/api/v1/ws/auctions`
- **Handshake Token Query:** `?token=` (opsional jika hanya menonton).

3.1 Format Frame Masuk (Inbound Frame dari Client ke Server)

```
{
  "action": "SUBSCRIBE_LOT",
  "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421"
}
```

3.2 Format Frame Siaran Lengkap (Outbound Events dari Server ke Client)

1. Event `PRICE_UPDATED` (Penawaran Baru Sah)
```

```
{
  "event": "PRICE_UPDATED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "current_price": 390000000.00,
    "highest_bidder_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "highest_bidder_username": "budi_kolektor",
    "increment_added": 10000000.00,
    "total_bids_count": 28
  }
}
```

2. Event `TIME_EXTENDED` (Anti-Sniping Aktif)

```
{
  "event": "TIME_EXTENDED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "extension_seconds": 30,
    "new_end_time": "2026-10-25T12:00:30.000Z",
    "reason": "ANTI_SNIPING_TRIGGERED"
  }
}
```

3. Event `AUCTION_BUYOUT_CLOSED` (Harga Maksimal Tersentuh)

```
{
  "event": "AUCTION_BUYOUT_CLOSED",
  "data": {
    "lot_id": "e0b9b3e1-872b-42e7-91a6-f7615cb38421",
    "winner_id": "3fa85f64-5717-4562-b3fc-2c963f66afa6",
    "winner_username": "budi_kolektor",
    "buyout_price": 500000000.00,
    "closed_reason": "VALUATION_CEILING_REACHED"
  }
}
```

4. Event `AUCTION_CONCLUDED` (Selesai Waktu & Settle)

```
{
  "event": "AUCTION_CONCLUDED",
  "data": {
    "lot_id": "c1f76d22-192a-4c28-8742-1928374aef12",
    "status": "SETTLED",
    "winner_id": "8bb30e92-3a52-47d1-ba76-218f2d5f22e1",
    "winner_username": "andi_collector",
    "final_price": 310000000.00,
    "collateral_unfrozen_for_others": true
  }
}
```

5. Event `NEW_LOT_PUBLISHED`

```
{
  "event": "NEW_LOT_PUBLISHED",
  "data": {
    "lot_id": "c1f76d22-192a-4c28-8742-1928374aef12",
    "title": "Hermès Birkin 25 Togo Gold Hardware",
    "brand": "Hermès",
    "category": "DESIGNER_BAGS",
    "starting_price": 250000000.00,
    "model_type": "CAPPED",
    "end_time": "2026-10-25T12:30:00.000Z"
  }
}
```

---

4. Dokumen Pengesahan (Document Sign-Off)

Spesifikasi kontrak antarmuka API (*API Contract Collection & Event Schemas*) ini telah ditinjau dan disetujui sebagai kontrak komunikasi baku antara tim Backend, Frontend Web (Next.js), dan Integrator Sistem.

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